

Removing all those terms that appear more than once, we finally obtain:

$$F = (x+y+z)(x+y'+z)(x+y+z)(x+y'+z) \\ = M_0 M_2 M_4 M_5$$

A convenient way to express this function is as follows: $F(x,y,z) = \prod(0,2,4,5)$

2.5.4 Conversion between Canonical Forms

Consider the function: $F(A,B,C) = \sum(1,4,5,6,7)$

This has a complement that can be expressed as: $F'(A,B,C) = \sum(0,2,3)$
 $= m_0 + m_2 + m_3$

Now, if we take the complement of F' by De Morgan's theorem, we obtain
 F in a different form: $F = (m_0 + m_2 + m_3)' = m_0' m_2' m_3' = M_0 M_2 M_3$
 $= \prod(0,2,3)$

From Table 2-3, the following relation holds true: $m_j' = M_j$.
 The maxterm with subscript j is a complement of the minterm with the same subscript j and vice-versa

General conversion procedure

To convert from one canonical form to another, interchange the symbols $\sum$ and $\prod$ and list those numbers missing from the original form.

$F(x,y,z) = \prod(0,2,4,5) \rightarrow$ product of maxterm form.
 $F'(x,y,z) = \sum(1,3,6,7) \rightarrow$ sum of minterm form.

2.5.5 Standard Forms

The sum of products is a Boolean expression containing AND terms, called product terms, of one or more literals each. The sum denotes the ORing of these terms. An example of a function expressed in sum of products is:

$$F_1 = y' + xz + x'yz'$$